

Xebia Internship Day 4

Assignment 3

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Q1) Bug 01 api_client.py

```
import requests
```

```
API_BASE_URL = "https://hr-internal.company.com/api"
```

```
API_TOKEN = "your_token_here"
```

```
def get_employee(employee_id):
```

```
    url = f"{API_BASE_URL}/employees/{employee_id}"
```

```
    # Fix Bug A: Added Authorization header with Bearer token.
```

```
    response = requests.get(
```

```
        url,
```

```
        headers={"Authorization": f"Bearer {API_TOKEN}"}
```

```
    )
```

```
    # Fix Bug B: Return None if employee is not found.
```

```
    if response.status_code == 404:
```

```
        return None
```

```
    data = response.json()
```

```
    return data["employee"]
```

Q2) Bug 02 db_queries.py

```
def get_monthly_attendance(conn, employee_id, month, year):  
    query = ""  
  
    # Fix Bug B: Passed employee_id in query parameters.  
    cursor.execute(query, (month, year, employee_id))  
  
    return cursor.fetchall()
```

Q3) Bug 03 attendance.py

```
from datetime import datetime
```

```
WORK_START = "09:00"
```

```
LATE_THRESHOLD_MINUTES = 15
```

```
def calculate_hours(check_in: str, check_out: str):
```

```
    fmt = "%H:%M"
```

```
    ci = datetime.strptime(check_in, fmt)
```

```
    co = datetime.strptime(check_out, fmt)
```

```
    # Fix Bug A: Subtract check-in from check-out to get correct duration.
```

```
    duration = co - ci
```

```
    hours_worked = duration.seconds / 3600
```

```
    start = datetime.strptime(WORK_START, fmt)
```

```
    late_by = (ci - start).seconds // 60
```

```
    # Fix Bug B: Use >= so exactly 15 minutes late is also flagged.
```

```
is_late = late_by >= LATE_THRESHOLD_MINUTES
```

```
return round(hours_worked, 2), is_late
```

Q4) Bug 04 report.py

```
def generate_summary(records):
```

```
    total_hours = 0
```

```
    late_days = 0
```

```
    # Fix Bug A:
```

```
    for i in range(len(records)):
```

```
        record = records[i]
```

```
        total_hours += record["hours"]
```

```
        if record["is_late"]:
```

```
            late_days += 1
```

```
    # Fix Bug B:
```

```
    avg_hours = total_hours / len(records) if len(records) > 0 else 0
```

```
    return {
```

```
        "total_hours": round(total_hours, 2),
```

```
        "avg_hours": round(avg_hours, 2),
```

```
        "late_days": late_days,
```

```
        "days_present": len(records)
```

```
    }
```

```
MIN_DAYS_REQUIRED = 20
```

```
MAX_LATE_DAYS = 3
```

Q5) Bug 05 report.py

```
def check_attendance_policy(summary):
```

```
    days_present = summary["days_present"]
```

```
    late_days = summary["late_days"]
```

```
    # Fix Bug A:
```

```
    below_minimum = days_present < MIN_DAYS_REQUIRED
```

Fix Bug B:

exceeded_late = late_days >= MAX_LATE_DAYS

if below_minimum or exceeded_late:

return {

"warning": True,

"reason": []

+ (["Below minimum attendance"] if below_minimum else [])

+ (["Exceeded late check-ins"] if exceeded_late else [])

}

return {"warning": False, "reason": []}